



SYLLABUS

SCHOOL OF ENGINEERING

SYLLABUS

FOR

PROGRAM: B.TECH.
[MECHANICAL ENGINEERING]

ACADEMIC YEAR: 2019 – 20
(BATCH: 2017-2021)

SEMESTER: V
[DEPARTMENT ELECTIVE – I]

DIRECTOR
SCHOOL OF ENGINEERING
RK UNIVERSITY
RAJKOT

SCHOOL OF ENGINEERING	PROGRAM: B. Tech (MECHANICAL ENGINEERING)
ACADEMIC YEAR - 2019-20	SEMESTER – V (Batch: 2017-21)
DEFINITION OF ONE CREDIT:	1. Lecture (L): 1 hour / week / semester, 2. Practical (P): 2 hour / week / semester, 3. Tutorial (T): 2 hour / week / semester.

DEPARTMENTAL ELECTIVE - I									
Course Code	Course Name	Teaching Hours			Credits	Audit course	CIE	PSEE	Remarks if any
		Theory	Tutorial	Practical					
ME517	INTRODUCTION TO AIRCRAFT INDUSTRY AND AIRCRAFT SYSTEMS	3	0	2	4	N	Y	Y	Offered Any One from Dept. Elective - I
ME518	CONTROL ENGINEERING	3	0	2	4	N	Y	Y	
ME520	TRIBOLOGY	3	0	2	4	N	Y	Y	
ME521	MECHATRONICS	3	0	2	4	N	Y	Y	
ME522	RELIABILITY ENGINEERING	3	0	2	4	N	Y	Y	
ME523	PRODUCT DESIGN	3	0	2	4	N	Y	Y	
	PROGRAMMING WITH PYTHON	3	0	2	4	N	Y	Y	

Date:

Director
Faculty of Technology
RK UNIVERSITY
Rajkot

DEPARTMENT ELECTIVE –I

Course Title	PRODUCT DESIGN (DEPT. ELECTIVE – I)	
Course Code	ME524	
Course Credit	Lecture : 03	
	Practical : 01	
	Tutorial : 00	
	Total : 04	
Course Objective		
After Successful completion of the above course, students will be able to: • Understand method of Design Thinking. • Implement Design Thinking process for the problem solving. • Understand the basic concepts of Product design with focus on the front-end processes. • Demonstrate the overview of all the product design processes and knowledge of concept generation. • Create a new product using Design Thinking and Product Design process with CAD tools. • Recognize Intellectual Property		
Detailed Syllabus		
Sr. No.	Name of chapter & Details	Hours Allotted
Section-I		
1	Design Thinking: What is design thinking? Functions and need of Design Thinking, Different phases of design thinking – Empathize, Define, Ideate, Prototype, Test.	02
2	Empathize Importance, Empathy building methods	02
3	Define Meaning and need of Define stage, Problem Statement, How to write meaningful problem statement.	02
4	Ideate What is ideation? Need and Importance, The process of ideation, Brain Storming, Mind Mapping, Role Play, Story Boarding	02
5	Prototype and Testing Need of Prototyping, Prototyping Methods, Need of Testing, Different Methods	01
6	Design thinking case studies and projects Case Studies on Design Thinking Projects on Design Thinking	12 06*
Total		21
Section-II		

7	Introduction to Product Design Product, Types of Product, Characteristics of successful Product, Duration and cost of Product development, The challenges of Product development, UX(User Experience) Design, Product design and Design thinking.	02
8	Sketching and Ideation Introduction, What is sketching, Need of sketching, Sketching and Ideation, Problem solving with sketching and ideation	06
9	Concept Generation Introduction, Need of Concept generation, Representing concepts with sketches, Case studies and examples, Aesthetic and Ergonomics, Design for Manufacturing, Design for Assembly	11
10.	Product Design using CAD. Application of CAD(2D + 3D) for product design, Case studies and examples.	06*
11	Intellectual Property (IP) Introduction, Need of IP, Trademark, Copyright, Patent Eligibility, Patent Distinctions, Provisional Patents, Patent Search, Working of Indian Patent Office	02
12	Design Project	06*
Total		21
* This hours will be counted towards laboratory hours		

Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 10 Marks weight age.
- Two internal exams will be conducted and each of the same will be converted to equivalent of 20 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weight age of 10 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 10 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the practical session.
- In the lectures and laboratory discipline and behaviour will be observed strictly.

Reference Books:

1. Vianna Mauricio, Ysmar Vianna, "Design Thinking Business Innovation", MJV Press
2. Soegaard Mads, "The Basics of User Experience Design", 1st edition, Interaction Design Foundation & bookboon.com
3. Parsons Tim, "Thinking: Objects Contemporary approaches to product design", AVA Publishing, Switzerland
4. Karl T. Ulrich, "Product design and development", Tata McGraw Hill.
5. Kevin Otto, Kristin Wood, "Product Design – Techniques in Reverse Engineering and New product Development", Pearson publication.
6. Haik Yousef, Shahin Tamer, "Engineering Design Process", Cengage Learning

7. George E. Dieter and Linda C. Schmidt, “Engineering Design”, Tata McGraw Hill

Additional Resources

- <https://dschool.stanford.edu/>
- <https://academy.autodesk.com/course/134629/tinkercad-fusion-360>
- <https://academy.autodesk.com/node/134554>
- <http://www.ulrich-eppinger.net/instructorpage.html>
- <https://www.creativemechanisms.com/blog/topic/design-cad>